

The Championship Gap: A Data Science Investigation into Real Madrid and Barcelona's UEFA Champions League Performance 2012–2018

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Abstract

This paper serves as the second installment of the football data analysis studies, presenting a forensic data science investigation into one of football's most compelling questions: how did Real Madrid claim four UEFA Champions League titles between 2012 and 2018 while Barcelona, home to arguably the greatest player in history, won only one? Both clubs were widely regarded as the finest teams in the world during this period, yet their championship outcomes diverged significantly. Rather than relying on emotional judgment or reputation, this paper applies objective data analysis to identify the measurable performance differences that explain the gap.

The paper examines each team's performance independently against their respective UEFA Champions League opponents across eleven modules: 1, Research background and the central question. 2, Data sources, scope, and key limitations. 3, The five-pillar analytical framework overview. 4-8, Analytical findings per pillar — shot analysis, passing analysis, danger pass, valued action, and defending respectively. 9, Tactical and structural synthesis explaining the performance gap. 10, Conclusions on key contributing factors. 11, Future plan introducing an AI agent to automate the framework for 2019–2024 UEFA Champions League seasons, building on the methodological foundation established in Parts 1 and 2.

Dedication

I would like to dedicate this paper to my favorite player Luka Modrić, who is approaching the twilight of an extraordinary career that has redefined what intelligence and grace mean in football. His brilliance on the pitch inspires me to conduct this set of studies to identify meaningful quantitative characteristics that explain the intelligence, quality, tactics and strategy of the game.

Acknowledgments

All references will be included in the paper. I would like to sincerely appreciate the work and education conducted by Jan Van Haaren, [Jan Van Haaren | LinkedIn](#), whose efforts in **building an evidence-based culture at Club Brugge** deeply resonate with the spirit of this study. As he noted, the most difficult challenge is not the analysis itself but changing the culture — convincing the industry to trust and invest in data over intuition and tradition. This paper is a step toward that same goal. I would also like to acknowledge Anthropic's Claude, whose AI tools significantly improved the efficiency of this study — Claude Chat supported the analytical thinking and interaction throughout the research process, while Claude Code accelerated the data analysis, programming, and visualization work through its strong programming capabilities.

Module 1 Research Background and the Central Question

The period between 2012 and 2018 represents one of the most compelling eras in the history of football. Two clubs, Real Madrid and Barcelona, maintained an extraordinary level of dominance simultaneously, concentrating an unprecedented density of world-class talent, visionary coaches, and tactical sophistication within the same league and the same European competition. While debates exist over when each club reached their absolute peak, few would dispute that this era placed both teams among the finest in the sport's history at the same time.

What makes this era particularly fascinating is the striking reversal of dominance across two different competitions. In La Liga, Barcelona was the superior force, claiming four titles between 2012 and 2018 compared to Real Madrid's one. Yet in the UEFA Champions League, the outcome was precisely the opposite — Real Madrid claimed four titles across the same period while Barcelona won only one. Both clubs were operating at the highest level simultaneously, yet their outcomes diverged completely depending on the stage.

Barcelona's domestic dominance during this era stands as a genuine testament to their quality, consistency, and tactical sophistication. Their La Liga record reflects a club in outstanding form across multiple seasons and should not be overlooked when evaluating their legacy. However, this paper does not conduct a domestic analysis — the scope is limited exclusively to UEFA Champions League performance, where the competition is knockout-based, opponent quality is at its highest, and the margin for error is significantly smaller.

The football industry has long operated on instinct, tradition, and narrative. Victories are celebrated and defeats are mourned, but the deeper structural and tactical reasons behind sustained success or underperformance are rarely examined with the rigor they deserve. If we truly want to honor the legacy of this golden era, we owe it to the sport to move beyond emotional judgment and understand what actually happened on the pitch, **quantitatively and objectively**.

This paper applies forensic data analysis to answer a simple but profound question: what **measurable performance differences** might explain why Real Madrid dominated the UEFA Champions League while Barcelona, equally gifted and domestically superior, could not replicate that consistency on the European stage?

Module 2 Data Sources, Scope, and Key Limitations

The match event data used in this study was sourced from WhoScored, which distributes data collected by Opta. The raw data was stored in JSON format, organized by competition, season, and individual match files. Each match file contains a structured list of events including passes, shots, duels, clearances, interceptions, and goalkeeper actions, alongside player and team identifiers, timestamps, and pitch coordinates.

Opta uses a normalized 100×100 coordinate system, where the entire pitch is represented as a perfect square regardless of actual pitch dimensions. While this approach allows consistent comparison across matches played on pitches of varying sizes, it treats length and width as equal, which does not reflect the true rectangular geometry of a football pitch.

All analytical scripts developed for this study were built on the StatsBomb data structure, which uses a 120×80 coordinate system that more accurately represents the realistic length-to-width ratio of a football pitch. StatsBomb was selected as the unified data standard for three reasons. First, it offers greater data granularity than Opta, with richer event attributes and more detailed field descriptions. Second, it is the industry standard for open football analytics, with all major analytical libraries such as mplsoccer built around its structure. Third, maintaining a single unified system across all studies ensures consistency and reusability of the analytical pipeline.

To bridge the two systems, a custom converter was developed to transform the Opta JSON data into a StatsBomb-compatible structure. The conversion process involved four key steps. First, all pitch coordinates were rescaled from the Opta 100×100 system to the StatsBomb 120×80 system. Second, Opta event type identifiers were mapped to their StatsBomb equivalents, covering passes, shots, duels, clearances, interceptions, fouls, goalkeeper actions, and substitutions. Third, nested qualifier fields in Opta — such as pass end location, pass length, pass angle, and shot outcome — were extracted and restructured to match StatsBomb field conventions. Fourth, numeric team and player identifiers were resolved to human-readable names using lookup tables built from the match data.

The converted data was then saved in Python pickle (.pkl) format rather than retained as JSON. This decision was made for three practical reasons: pkl files load significantly faster than JSON for large datasets, they preserve Python data types such as DataFrames and dictionaries without requiring re-parsing, and they reduce the overhead of repeated file reading during iterative analysis.

It is important to acknowledge two limitations arising from this conversion. The first is coordinate distortion. Rescaling from a 100×100 square system to a 120×80 rectangular system changes the aspect ratio of the pitch, meaning that absolute distances and angles computed from the converted data carry a degree of geometric distortion. Pass lengths, shot distances, and zone boundary positions are all affected. The second limitation is field incompleteness. StatsBomb's native data includes fields such as expected goals (xG) and freeze frame snapshots that have no equivalent in the Opta dataset and therefore cannot be populated through conversion.

However, both Real Madrid and Barcelona datasets underwent identical conversion treatment. The distortion is therefore systematic and consistent across both clubs. Since the purpose of this study is comparative rather than absolute, the relative differences between the two clubs remain analytically valid. The conclusions drawn in this paper are based on performance gaps between the two teams, not on absolute metric values, and the uniform treatment of both datasets preserves the integrity of that comparison.

Table 1 summarizes the key structural differences between the original Opta data and the converted StatsBomb-compatible data used in this study.

Table 1 The difference of original WhoScored and StabsBomb Compatible Data

Aspect	Opta / WhoScored	Converted StatsBomb-Compatible
Coordinate System	100×100 (square, normalized)	120×80 (rectangular, realistic ratio)
Storage Format	JSON	pkl (Python pickle)
Event Type Reference	Numeric type IDs	Named type strings
Location Fields	x, y as flat values	x, y, end_x, end_y as split columns
Qualifier Fields	Nested list of key-value pairs	Extracted and flattened
Player / Team Reference	Numeric IDs	Resolved to names via lookup
xG Field	Not available	Not available (conversion limitation)
Freeze Frame Data	Not available	Not available (conversion limitation)

The study covers six UEFA Champions League seasons from 2012-2013 to 2017-2018, encompassing a total of 142 matches — 76 for Real Madrid and 66 for Barcelona. Each club's matches are recorded independently against their respective opponents across all competition stages, from the Group Stage through to the Final. Tables 2 through 7 present the complete match records for both clubs by season, including the date, home or away designation, opponent, score, result, and competition round. This full transparency of the dataset allows readers to understand the scope and scale of the analysis, and to verify the context of any specific match referenced in the subsequent analytical modules.

Table 2 UEFA Champions League Match Records — Season 2012-2013

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2012-09-18	H	Man City	3 : 2	W	Group Stage MD1
Barcelona	2012-09-19	H	Spartak Moscow	3 : 2	W	Group Stage MD1
Barcelona	2012-10-02	A	Benfica	0 : 2	W	Group Stage MD2
Real Madrid	2012-10-03	A	Ajax	1 : 4	W	Group Stage MD2
Barcelona	2012-10-23	H	Celtic	2 : 1	W	Group Stage MD3
Real Madrid	2012-10-24	A	Borussia Dortmund	2 : 1	L	Group Stage MD3
Real Madrid	2012-11-06	H	Borussia Dortmund	2 : 2	D	Group Stage MD4
Barcelona	2012-11-07	A	Celtic	2 : 1	L	Group Stage MD4
Barcelona	2012-11-20	A	Spartak Moscow	0 : 3	W	Group Stage MD5
Real Madrid	2012-11-21	A	Man City	1 : 1	D	Group Stage MD5
Real Madrid	2012-12-04	H	Ajax	4 : 1	W	Group Stage MD6
Barcelona	2012-12-05	H	Benfica	0 : 0	D	Group Stage MD6
Real Madrid	2013-02-13	H	Man Utd	1 : 1	D	Round of 16 Leg 1
Barcelona	2013-02-20	A	AC Milan	2 : 0	L	Round of 16 Leg 1
Real Madrid	2013-03-05	A	Man Utd	1 : 2	W	Round of 16 Leg 2

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2013-03-12	H	AC Milan	4 : 0	W	Round of 16 Leg 2
Barcelona	2013-04-02	A	PSG	2 : 2	D	Quarter-Final Leg 1
Real Madrid	2013-04-03	H	Galatasaray	3 : 0	W	Quarter-Final Leg 1
Real Madrid	2013-04-09	A	Galatasaray	3 : 2	L	Quarter-Final Leg 2
Barcelona	2013-04-10	H	PSG	1 : 1	D	Quarter-Final Leg 2
Barcelona	2013-04-23	A	Bayern	4 : 0	L	Semi-Final Leg 1
Real Madrid	2013-04-24	A	Borussia Dortmund	4 : 1	L	Semi-Final Leg 1
Real Madrid	2013-04-30	H	Borussia Dortmund	2 : 0	W	Semi-Final Leg 2
Barcelona	2013-05-01	H	Bayern	0 : 3	L	Semi-Final Leg 2

Table 3 UEFA Champions League Match Records — Season 2013-2014

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2013-09-17	A	Galatasaray	1 : 6	W	Group Stage MD1
Barcelona	2013-09-18	H	Ajax	4 : 0	W	Group Stage MD1
Barcelona	2013-10-01	A	Celtic	0 : 1	W	Group Stage MD2
Real Madrid	2013-10-02	H	Copenhagen	4 : 0	W	Group Stage MD2
Barcelona	2013-10-22	A	AC Milan	1 : 1	D	Group Stage MD3
Real Madrid	2013-10-23	H	Juventus	2 : 1	W	Group Stage MD3
Real Madrid	2013-11-05	A	Juventus	2 : 2	D	Group Stage MD4
Barcelona	2013-11-06	H	AC Milan	3 : 1	W	Group Stage MD4
Barcelona	2013-11-26	A	Ajax	2 : 1	L	Group Stage MD5

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2013-11-27	H	Galatasaray	4 : 1	W	Group Stage MD5
Real Madrid	2013-12-10	A	Copenhagen	0 : 2	W	Group Stage MD6
Barcelona	2013-12-11	H	Celtic	6 : 1	W	Group Stage MD6
Barcelona	2014-02-18	A	Man City	0 : 2	W	Round of 16 Leg 1
Real Madrid	2014-02-26	A	Schalke	1 : 6	W	Round of 16 Leg 1
Barcelona	2014-03-12	H	Man City	2 : 1	W	Round of 16 Leg 2
Real Madrid	2014-03-18	H	Schalke	3 : 1	W	Round of 16 Leg 2
Barcelona	2014-04-01	H	Atletico	1 : 1	D	Quarter-Final Leg 1
Real Madrid	2014-04-02	H	Borussia Dortmund	3 : 0	W	Quarter-Final Leg 1
Real Madrid	2014-04-08	A	Borussia Dortmund	2 : 0	L	Quarter-Final Leg 2
Barcelona	2014-04-09	A	Atletico	1 : 0	L	Quarter-Final Leg 2
Real Madrid	2014-04-23	H	Bayern	1 : 0	W	Semi-Final Leg 1
Real Madrid	2014-04-29	A	Bayern	0 : 4	W	Semi-Final Leg 2
Real Madrid	2014-05-24	H	Atletico	4 : 1 (AET)	W	Final

Table 4 UEFA Champions League Match Records — Season 2014-2015

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2014-09-16	H	FC Basel 1893	5 : 1	W	Group Stage MD1
Barcelona	2014-09-17	H	APOEL Nicosia	1 : 0	W	Group Stage MD1
Barcelona	2014-09-30	A	PSG	3 : 2	L	Group Stage MD2
Real Madrid	2014-10-01	A	Ludogorets Razgrad	1 : 2	W	Group Stage MD2

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2014-10-21	H	Ajax	3 : 1	W	Group Stage MD3
Real Madrid	2014-10-22	A	Liverpool	0 : 3	W	Group Stage MD3
Real Madrid	2014-11-04	H	Liverpool	1 : 0	W	Group Stage MD4
Barcelona	2014-11-05	A	Ajax	0 : 2	W	Group Stage MD4
Barcelona	2014-11-25	A	APOEL Nicosia	0 : 4	W	Group Stage MD5
Real Madrid	2014-11-26	A	FC Basel 1893	0 : 1	W	Group Stage MD5
Real Madrid	2014-12-09	H	Ludogorets Razgrad	4 : 0	W	Group Stage MD6
Barcelona	2014-12-10	H	PSG	3 : 1	W	Group Stage MD6
Real Madrid	2015-02-18	A	Schalke	0 : 2	W	Round of 16 Leg 1
Barcelona	2015-02-24	A	Man City	1 : 2	W	Round of 16 Leg 1
Real Madrid	2015-03-10	H	Schalke	3 : 4	L	Round of 16 Leg 2
Barcelona	2015-03-18	H	Man City	1 : 0	W	Round of 16 Leg 2
Real Madrid	2015-04-14	A	Atletico	0 : 0	D	Quarter-Final Leg 1
Barcelona	2015-04-15	A	PSG	1 : 3	W	Quarter-Final Leg 1
Barcelona	2015-04-21	H	PSG	2 : 0	W	Quarter-Final Leg 2
Real Madrid	2015-04-22	H	Atletico	1 : 0	W	Quarter-Final Leg 2
Real Madrid	2015-05-05	A	Juventus	2 : 1	L	Semi-Final Leg 1
Barcelona	2015-05-06	H	Bayern	3 : 0	W	Semi-Final Leg 1
Barcelona	2015-05-12	A	Bayern	3 : 2	L	Semi-Final Leg 2
Real Madrid	2015-05-13	H	Juventus	1 : 1	D	Semi-Final Leg 2
Barcelona	2015-06-06	A	Juventus	1 : 3	W	Final

Table 5 UEFA Champions League Match Records — Season 2015-2016

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2015-09-15	H	Shakhtar	4 : 0	W	Group Stage MD1
Barcelona	2015-09-16	A	Roma	1 : 1	D	Group Stage MD1
Barcelona	2015-09-29	H	Leverkusen	2 : 1	W	Group Stage MD2
Real Madrid	2015-09-30	A	Malmoe FF	0 : 2	W	Group Stage MD2
Barcelona	2015-10-20	A	BATE	0 : 2	W	Group Stage MD3
Real Madrid	2015-10-21	A	PSG	0 : 0	D	Group Stage MD3
Real Madrid	2015-11-03	H	PSG	1 : 0	W	Group Stage MD4
Barcelona	2015-11-04	H	BATE	3 : 0	W	Group Stage MD4
Barcelona	2015-11-24	H	Roma	6 : 1	W	Group Stage MD5
Real Madrid	2015-11-25	A	Shakhtar	3 : 4	W	Group Stage MD5
Real Madrid	2015-12-08	H	Malmoe FF	8 : 0	W	Group Stage MD6
Barcelona	2015-12-09	A	Leverkusen	1 : 1	D	Group Stage MD6
Real Madrid	2016-02-17	A	Roma	0 : 2	W	Round of 16 Leg 1
Barcelona	2016-02-23	A	Arsenal	0 : 2	W	Round of 16 Leg 1
Real Madrid	2016-03-08	H	Roma	2 : 0	W	Round of 16 Leg 2
Barcelona	2016-03-16	H	Arsenal	3 : 1	W	Round of 16 Leg 2
Barcelona	2016-04-05	H	Atletico	2 : 1	W	Quarter-Final Leg 1
Real Madrid	2016-04-06	A	Wolfsburg	2 : 0	L	Quarter-Final Leg 1
Real Madrid	2016-04-12	H	Wolfsburg	3 : 0	W	Quarter-Final Leg 2
Barcelona	2016-04-13	A	Atletico	2 : 0	L	Quarter-Final Leg 2
Real Madrid	2016-04-26	A	Man City	0 : 0	D	Semi-Final Leg 1

Team	Date	H/A	Opponent	Score	Result	Round
Real Madrid	2016-05-04	H	Man City	1 : 0	W	Semi-Final Leg 2
Real Madrid	2016-05-28	H	Atletico	1 : 1 (AET)	D	Final

Table 6 UEFA Champions League Match Records — Season 2016-2017

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2016-09-13	H	Celtic	7 : 0	W	Group Stage MD1
Real Madrid	2016-09-14	H	Sporting	2 : 1	W	Group Stage MD1
Real Madrid	2016-09-27	A	Borussia Dortmund	2 : 2	D	Group Stage MD2
Barcelona	2016-09-28	A	Borussia M.Gladbach	1 : 2	W	Group Stage MD2
Real Madrid	2016-10-18	H	Legia Warszawa	5 : 1	W	Group Stage MD3
Barcelona	2016-10-19	H	Man City	4 : 0	W	Group Stage MD3
Barcelona	2016-11-01	A	Man City	3 : 1	L	Group Stage MD4
Real Madrid	2016-11-02	A	Legia Warszawa	3 : 3	D	Group Stage MD4
Real Madrid	2016-11-22	A	Sporting	1 : 2	W	Group Stage MD5
Barcelona	2016-11-23	A	Celtic	0 : 2	W	Group Stage MD5
Barcelona	2016-12-06	H	Borussia M.Gladbach	4 : 0	W	Group Stage MD6
Real Madrid	2016-12-07	H	Borussia Dortmund	2 : 2	D	Group Stage MD6
Barcelona	2017-02-14	A	PSG	4 : 0	L	Round of 16 Leg 1
Real Madrid	2017-02-15	H	Napoli	3 : 1	W	Round of 16 Leg 1
Real Madrid	2017-03-07	A	Napoli	1 : 3	W	Round of 16 Leg 2

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2017-03-08	H	PSG	6 : 1	W	Round of 16 Leg 2
Barcelona	2017-04-11	A	Juventus	3 : 0	L	Quarter-Final Leg 1
Real Madrid	2017-04-12	A	Bayern	1 : 2	W	Quarter-Final Leg 1
Real Madrid	2017-04-18	H	Bayern	4 : 2 (AET)	W	Quarter-Final Leg 2
Barcelona	2017-04-19	H	Juventus	0 : 0	D	Quarter-Final Leg 2
Real Madrid	2017-05-02	H	Atletico	3 : 0	W	Semi-Final Leg 1
Real Madrid	2017-05-10	A	Atletico	2 : 1	L	Semi-Final Leg 2
Real Madrid	2017-06-03	A	Juventus	1 : 4	W	Final

Table 7 UEFA Champions League Match Records — Season 2017-2018

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2017-09-12	H	Juventus	3 : 0	W	Group Stage MD1
Real Madrid	2017-09-13	H	APOEL Nicosia	3 : 0	W	Group Stage MD1
Real Madrid	2017-09-26	A	Borussia Dortmund	1 : 3	W	Group Stage MD2
Barcelona	2017-09-27	A	Sporting	0 : 1	W	Group Stage MD2
Real Madrid	2017-10-17	H	Tottenham	1 : 1	D	Group Stage MD3
Barcelona	2017-10-18	H	Olympiacos	3 : 1	W	Group Stage MD3
Barcelona	2017-10-31	A	Olympiacos	0 : 0	D	Group Stage MD4
Real Madrid	2017-11-01	A	Tottenham	3 : 1	L	Group Stage MD4
Real Madrid	2017-11-21	A	APOEL Nicosia	0 : 6	W	Group Stage MD5
Barcelona	2017-11-22	A	Juventus	0 : 0	D	Group Stage MD5

Team	Date	H/A	Opponent	Score	Result	Round
Barcelona	2017-12-05	H	Sporting	2 : 0	W	Group Stage MD6
Real Madrid	2017-12-06	H	Borussia Dortmund	3 : 2	W	Group Stage MD6
Real Madrid	2018-02-14	H	PSG	3 : 1	W	Round of 16 Leg 1
Barcelona	2018-02-20	A	Chelsea	1 : 1	D	Round of 16 Leg 1
Real Madrid	2018-03-06	A	PSG	1 : 2	W	Round of 16 Leg 2
Barcelona	2018-03-14	H	Chelsea	3 : 0	W	Round of 16 Leg 2
Real Madrid	2018-04-03	A	Juventus	0 : 3	W	Quarter-Final Leg 1
Barcelona	2018-04-04	H	Roma	4 : 1	W	Quarter-Final Leg 1
Barcelona	2018-04-10	A	Roma	3 : 0	L	Quarter-Final Leg 2
Real Madrid	2018-04-11	H	Juventus	1 : 3	L	Quarter-Final Leg 2
Real Madrid	2018-04-25	A	Bayern	1 : 2	W	Semi-Final Leg 1
Real Madrid	2018-05-01	H	Bayern	2 : 2	D	Semi-Final Leg 2
Real Madrid	2018-05-26	H	Liverpool	3 : 1	W	Final

Module 3 Five-Pillar Analytical Framework

Understanding why one elite team consistently outperforms another in a knockout competition requires more than observing the final scores. It requires identifying the underlying elements of match performance that can be observed, recorded, and measured quantitatively. This study proposes that meaningful football analysis must begin with defining what to measure before any deeper quality control methodology can be applied. The five pillars presented in this module represent the author's attempt to identify the key measurable elements that may explain the performance gap between Real Madrid and Barcelona across six UEFA Champions League seasons.

The first pillar is Shot Analysis. Shots are the most direct measurable link to the outcome of a football match. Examining shot volume, shot location, shot type, and outcome provides the clearest quantitative window into a team's attacking effectiveness and finishing consistency. Understanding how and from where each team generated their shots — and how efficiently those shots were converted — is the most fundamental starting point for any performance evaluation.

The second pillar is Passing Analysis. Passing is the primary mechanism through which a team builds attacking sequences and maintains possession. Analyzing passing volume, passing accuracy, passing direction, and passing zones reveals how each team structured their play and whether their ball circulation was purposeful and progressive or lateral and conservative.

The third pillar is Danger Pass Analysis. A danger pass is defined in this study as a pass that directly creates a shooting opportunity. It serves as the analytical bridge between passing structure and shot generation, capturing the moments where passing quality translates into genuine attacking threat. This pillar connects the second and first pillars, helping to identify which passing patterns were most effective in creating high-quality chances.

The fourth pillar is Valued Action Analysis. While the first three pillars focus on the attacking chain, Valued Action takes a broader view of the full match, assigning measurable value to the range of actions performed by each team across all phases of play. This pillar is designed to reveal the style fingerprint of each club — how they generated value across different match situations — and provides context for interpreting the patterns observed in the other pillars.

The fifth pillar is Defending Analysis. Defense is not merely the absence of conceding goals. It is an active set of measurable behaviors including interceptions, clearances, defensive duels, and the spatial organization of the defensive unit. This pillar examines how each team suppressed their opponents' attacking opportunities, which is essential for understanding the full picture of match performance.

Together, these five pillars form a structured analytical lens through which the performance of Real Madrid and Barcelona can be examined objectively and comparatively. It is important to acknowledge that this framework represents an early stage of a longer analytical journey. Identifying these elements is the necessary first step. Developing robust methods to measure them with greater precision, and eventually understanding how to improve them systematically, remains the work ahead. This study contributes to that foundation.

To be continued on Module 4-6